ENGR 1113

Introduction to Engineering

Lipscomb University · Fall 2026

WillSpeak

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Scope & limits

Speakers: three public TORGO speakers with severe dysarthria (M01, M02, M04), reading prompts.

Sample size: 513 held-out clips (120 sentences, 393 single words).

01 Objective

PROBLEM

The client has cerebral palsy. People close to him still miss words, and strangers struggle to understand him. He is losing function with age.

OBJECTIVE

Hear him and repeat his words in a clear voice, with no buttons.

IMPACT

The client first, then others with dysarthria.



Fig. 1. The 3D-printed prototype.

TAKEAWAY

No hands, clear speech.

02 Constraints

WHAT SHAPED IT

Each fact set a requirement, shown in blue.

Off-the-shelf Parakeet-TDT 0.6B gets **59.6%** of words wrong on 292 severe TORGO sentences (95% CI 55–64%). Voiceitt works best on mild to moderate speech.

A model personalised to his voice

He cannot press a switch, button or screen.

Voice-only control: speak, stop, sleep, wake

It rides on his chair all day; idle power sets battery life.

Small, low-power and battery-run

A mic needs to not be in his face and a live test picked up other voices. At 20 cm, a collar mic gives him about 14 dB over a voice at 1 m.

A collar mic and a gate set to his speaking level

Other TORGO speakers read at least 98% of each severe speaker's prompts (Hui et al., 2024), a known issue. Prompts are split 70/10/20, the same for every speaker.

Test only on sentences kept out of training

TAKEAWAY

5 must-haves.

03 Design

HARDWARE

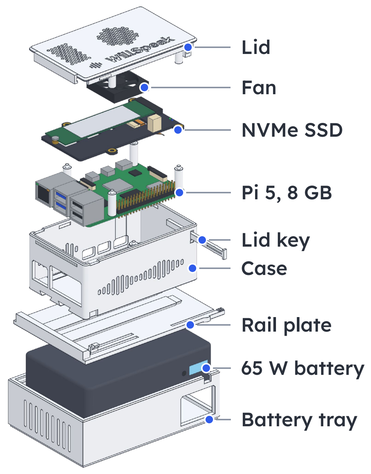


Fig. 2. Exploded view. White parts are 3D-printed.

Sensor	Lavalier mic
Actuator	Speaker

CODE FLOW

Microphone
raw audio, auto-gain off
↓
Loudness gate
12 dB under his median level
↓
Clean up
trims silence, skips duplicates
↓
Speech recognition
8-bit Parakeet-TDT 0.6B on Pi 5
↓
Language model
Qwen2.5-0.5B picks the best fit
↓
Hands-free check
speak it, stop it, or sleep
↓
Speaker
says it in a clear voice

TRAINING

1. **Stock model:** 618 M parameters
2. **Pretrain:** other severe speakers
3. **Adapter:** 1 M parameters fitted to the speaker's clips, rest frozen
4. **Export:** 8-bit file for the Pi 5

TAKEAWAY

Cheap, effective and widely accessible.

04 Testing & Results

TRAINING DATA

TORGO: 8 speakers with dysarthria reading words and sentences.

3 severe speakers (M01, M02, M04), 72–156 training clips each.

WHOLE SENTENCES RIGHT

120 held-out sentences, 3 speakers

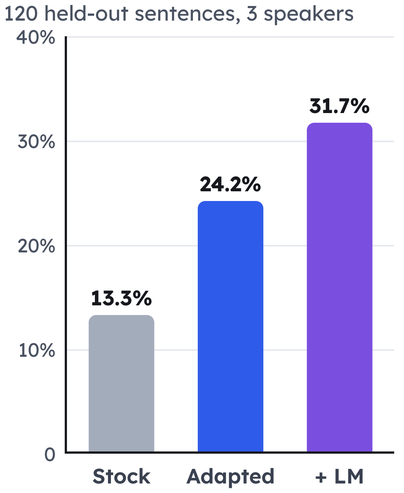


Fig. 3. Adapted: Training steps 2–3. LM: language model, +7.5 points (95% CI 2.5–12.5).

28.8% word error rate on 1,304 words (stock 68.1%)

0.40 s per 3 s phrase (Pi 5, stock)

WHAT TESTING CHANGED

Mini-PC too slow
↳ **Pi 5, 3.9× faster**

First fine-tune lost to stock
↳ **froze batch norm**

Live test heard other voices
↳ **loudness gate**

LM cost single words
↳ **own word chooser**

TAKEAWAY

Promising results, but limited by data.

05 The Future

PLAN

1. Record him & train his model
2. Mount it on his chair
3. Add a voice-identity gate

SUSTAINABILITY

Keeps learning as his speech changes with age.

Weekly overnight update: encrypted recordings retrain his model, then are deleted.

Daily use stays offline.

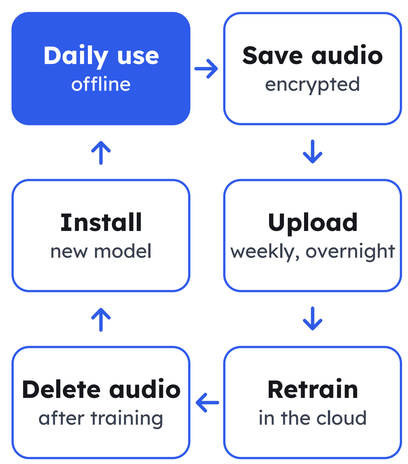


Fig. 4. Potential update system.

TAKEAWAY

Collect data, and improve.

Citations

Tool use: Claude (Anthropic), general assistant and tutor.

Rudzicz, Namasivayam & Wolff, *TORGO*, Language Resources and Evaluation 46(4), 2012

Hui, Zhang & Mohan, prompt overlap in TORGO, arXiv:2411.00980 (2024)

NVIDIA, Parakeet-TDT-0.6B-v2 speech model

Qwen team, Qwen2.5-0.5B language model

Scan for more.
rednetty.github.io/willspeak-site
3D model, results, code

Acknowledgements

Dr. Juan Rojas, ENGR 1113 professor